One-variable data: Distributions and measures of center and spread answer key

#	ID	Answer	Difficulty
1	a214fab5-1eb4-49e2-ad3e-2b74eb180945	C	Easy
2	862ae133-b6b4-494f-a4f0-2a3ec7706432	14	Easy
3	031602-DC	B	Easy
4	1bb852f0-0834-410f-84fd-8636ef74788b	D	Easy
5	32c749f0-91ae-4721-8ecd-9eac7aa1f372	B	Easy
6	43a2c649-5ad6-4d16-9779-da333ff1569a	B	Easy
7	b00de09c-a654-4119-9b48-e0de375ac094	A	Easy
8	f5598292-13c4-4d60-8b84-b381d3c86f9c	D	Easy
9	db1f16a5-024c-473c-aedc-cdda8967d1c1	79	Easy
10	88659c2c-a185-4a2b-b844-b4bdc689c709	8	Easy
11	5028e152-9c24-4fde-a87c-5eb62669957e	A	Easy
12	abb8d90e-7d43-41be-9dae-ad5e692240f5	B	Easy
13	da3d86e5-c459-4887-9ed0-fecf17d87773	C	Easy
14	012a45d5-4b75-4217-b792-5b7d08ece247	B	Easy
15	1010b663-23d7-48e5-8a54-02a9c7c6a552	B	Easy
16	028959-DC	B	Easy
17	a6ce86a7-db0b-4ece-b1dc-2fb1ebd4b99b	40	Easy
18	08099-DC	D	Easy
19	070909-DC	C	Easy
20	bc57ca71-eb48-4ae7-95a8-3ed00f622ed8	9	Easy

21	47edb639-8ebe-471a-ae9d-172195289d2b	B	Easy
22	0d175345-0c08-4bec-811b-d117627c8b48	6	Easy
23	027874-DC	D	Easy
24	026304-DC	C	Easy
25	132dc872-80cb-43fa-b1c9-3dc7fea2a1a5	B	Easy
26	ede10369-b903-44c6-9ea8-c55232a55dfd	30	Easy
27	ad801a2d-5e09-4c92-b37a-9c8b027d62f2	D	Easy
28	070910-DC	B	Easy
29	ee5e0bcd-6e26-4cc7-9ca8-16eab11e3146	B	Easy
30	a966fc8b-7920-4a60-ae82-b53184a3ba6c	A	Easy
31	070911-DC	B	Easy
32	06926-DC	A	Easy
33	8183ef2c-57c1-4464-91c3-475ac4162ca1	C	Easy
34	a76daad9-3856-4e79-95ee-20158581002b	C	Easy
35	026380-DC	B	Easy
36	09906-DC	A	Easy
37	11e14767-1eaf-4adb-8d85-9e42946ce439	B	Easy
38	14957-DC	C	Medium
39	4c64fb61-5036-48e2-a518-1fe34e097aec	10	Medium
40	11167-DC	A	Medium
41	020981-DC	B	Medium
42	10941-DC	C	Medium
43	070625-DC	?	Medium
44	029308-DC	B	Medium
45	7d86a5e3-51f9-4e0f-9b63-3de9936e8dd1	A	Medium
46	ee670944-e520-4f88-880a-a0158fe35ef8	19	Medium

47	a1b19718-6d73-483b-ac58-8fda7ec4a928	A	Medium
48	028792-DC	?	Medium
49	070626-DC	?	Medium
50	026381-DC	A	Medium
51	cc8606cf-9fd8-4c61-9cb6-1bcb5b969d36	B	Medium
52	2466799c-cc68-42e1-b0cb-461b4ca5597c	3630	Medium
53	d2d90742-b90d-4d9c-b0ec-3254255f86be	14	Medium
54	14851-DC	C	Medium
55	029327-DC	A	Medium
56	12877eac-b005-471d-851e-37b10d8a7f90	D	Medium
57	5af384de-ab0c-40e8-8d4d-b12e04a08e8e	B	Medium
58	e556a06b-805c-4238-8c14-3bcafe7e1ddf	29	Medium
59	00566-DC	C	Hard
60	b09d507c-3dde-4f24-af93-e07719f875cf	44	Hard
61	07831-DC	D	Hard
62	213dee0e-d9eb-4743-be96-e47241c0fb36	4	Hard
63	07857-DC	B	Hard
64	070627-DC	?	Hard
65	1f3a2144-9cce-41ef-bcb2-8dfc5cc97d8b	52	Hard
66	4f4352ff-e0b0-4c52-b1d8-8461b2e5eb6e	B	Hard
67	4135b77c-4178-4cdc-8f4a-8f46e634b934	D	Hard
68	028957-DC	D	Hard
69	d7e43cb0-447a-4672-bf3c-3e744dd7843f	B	Hard
70	029610-DC	A	Hard
71	b9761b18-7942-4214-89a5-1726fe26fe75	41	Hard
72	f1db57ba-417f-4c96-ae7e-80f171f39606	C	Hard

73	070912-DC	?	Hard
74	32297287-d739-4961-acc3-2eed761e3e0d	A	Hard
75	a97f29fd-deb9-4f76-baff-4f4bea70e929	B	Hard
76	45c46b40-e7e6-4f3e-bbb1-22e0862c3911	A	Hard
77	f98d0c41-892c-4afb-82e0-d348eba42052	C	Hard
78	16b20598-30a0-44b7-8771-8c87e56ec14b	C	Hard
79	bb8df1b9-8ab9-42bb-8b7f-adec3639c7f8	A	Hard
80	46b54ee4-c738-4f84-acf0-ed71e853d400	B	Hard
81	84fea272-2bed-48c4-9ffd-dfe5cc5edb2f	C	Hard
82	27d95f70-58ff-4b21-92f7-c74df5011b3a	D	Hard
83	33c49cb1-9a61-48d8-8b47-83909effa87d	C	Hard
84	a4fc06ae-2b85-4830-a27c-e248eebde1f9	B	Hard

Rationales

▼ 1 — C

Choice C is correct. It's given that the numbers of piglets in litters 1, 2, and 3 are 8, 7, and 9, respectively. The mean number of piglets per litter for these three litters is $\frac{8+7+9}{3}$, which is equivalent to $\frac{24}{3}$, or 8.

Choice A is incorrect. This is the range of the number of piglets per litter for these three litters.

Choice B is incorrect. This is the minimum number of piglets per litter for these three litters.

Choice D is incorrect. This is the maximum number of piglets per litter for these three litters.

▼ 2 — 14

The correct answer is 14. The table summarizes the distribution of shape printed on one side for cards in a deck of cards. The table shows that the number of cards when the shape is a triangle is 14. Therefore, there are 14 cards in the deck that have a triangle printed on one side.

▼ 3 — B

Choice B is correct. The mean land area of these 5 countries is equal to the sum of the land areas of these countries, or $2.0+21+61+26+0.44$, divided by the number of countries in the table, 5, or $\frac{2.0+21+61+26+0.44}{5}$. Combining like terms in the numerator yields $\frac{110.44}{5}$, which simplifies to 22.088 square kilometers. This value, when rounded to the nearest square kilometer, is 22.

Choice A is incorrect and may result from a calculation error. Choice C is incorrect. This is the greatest land area of the 5 countries in the table. Choice D is incorrect. This is the sum of the land areas of the 5 countries in the table, rounded to the nearest square kilometer.

▼ 4 — D

Choice D is correct. The height of each bar in the graph shown represents the number of volunteers who chose the gift labeled with the letter specified at the bottom of the bar. The bar for gift C has a height of 18. Therefore, 18 volunteers chose gift C.

Choice A is incorrect. This is the number of volunteers who chose gift D, not gift C.

Choice B is incorrect. This is the number of volunteers who chose gift B, not gift C.

Choice C is incorrect. This is the number of volunteers who chose gift A, not gift C.

▼ 5 — B

Choice B is correct. The dot plot gives the diameter, to the nearest inch, of each sea star in a group of tide pools. The number of dots above a given diameter represents the number of sea stars with that diameter, to the nearest inch. There are 6 dots above 16 inches. Therefore, 6 sea stars had a diameter, to the nearest inch, of 16 inches.

Choice A is incorrect and may result from calculation or conceptual errors.

Choice C is incorrect. This is the number of sea stars with a diameter, to the nearest inch, of 17 inches, not 16 inches.

Choice D is incorrect. This is the number of sea stars with a diameter, to the nearest inch, of 18 inches, not 16 inches.

▼ 6 — B

Choice B is correct. The mean of a set of data values is the sum of all the data values divided by the number of data values in the set. The sum of the data values shown is $2 + 9 + 14 + 23 + 32$, or 80. Since there are 5 data values in the set, the mean of the data shown is $\frac{80}{5}$, or 16.

Choice A is incorrect. This is the median, not the mean, of the data shown.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the maximum, not the mean, of the data shown.

▼ 7 — A

Choice A is correct. It's given that the bar graph shows the distribution of the number of students in each of four extracurricular activities at a high school. The bar representing drama has a height of 40; therefore, there are 40 students in drama. The bar representing chess has a height of 30; therefore, there are 30 students in chess. Thus, there are $40 - 30$, or 10 more students in drama than in chess.

Choice B is incorrect. This is the number of students in chess.

Choice C is incorrect. This is the number of students in drama.

Choice D is incorrect. This is the sum of the number of students in drama and in chess.

▼ 8 — D

Choice D is correct. It's given that the bar graph summarizes the charge, in kilowatt-hours kWh, a battery received each day for 15 days. The height of each bar in the bar graph shown represents the number of days the battery received the charge, in kWh, specified at the bottom of the bar. The bar for a charge of $0 \ \text{kWh}$ reaches a height of 6. Therefore, the battery received a charge of $0 \ \text{kWh}$ for 6 of these days.

Choice A is incorrect. This is the charge, in kWh, that the battery received, not the number of days the battery received this charge.

Choice B is incorrect. This is the number of days the battery received a charge of either 8, 16, or 23 kWh.

Choice C is incorrect. This is the number of days the battery received a charge of 11 kWh.

▼ 9 — 79

The correct answer is 79. The median of a data set with an odd number of values is the middle value of the set when the values are ordered from least to greatest. Because the given data set consists of nine values that are ordered from least to greatest, the median is the fifth value in the data set. Therefore, the median of the data shown is 79.

▼ 10 — 8

The correct answer is 8. The mean of a data set is the sum of the values in the data set divided by the number of values in the data set. It's given in the table that the times, in minutes, it took Robert to complete each of the tasks are 5, 2, 15, 9, and 9. The sum of these times is $5 + 2 + 15 + 9 + 9$, or 40, minutes. There are 5 tasks in the project. Therefore, the mean time, in minutes, for Robert to complete a task for this project is $40 \div 5$, or 8.

▼ 11 — A

Choice A is correct. A frequency table is a table that lists the data value and shows the number of times the data value occurs. In the data listed, the number 4 occurs four times, the number 8 occurs three times, and the number 13 occurs two times. This corresponds to the table in choice A.

Choice B is incorrect. This table has the values for number and frequency reversed.

Choice C is incorrect because the frequency values don't represent the data listed.

Choice D is incorrect. This table represents the listed number values as the frequency values.

▼ 12 — B

Choice B is correct. The mean of a data set is the sum of the values in the data set divided by the number of values in the data set. It follows that the mean of data set X is $\frac{5 + 9 + 9 + 13}{4}$,

or 9, and the mean of data set Y is $\frac{5+9+9+13+27}{5}$, or 12.6. Since 9 is less than 12.6, the mean of data set X is less than the mean of data set Y.

Alternate approach: Data set Y consists of the 4 values in data set X and one additional value, 27. Since the additional value, 27, is larger than any value in data set X, the mean of data set X is less than the mean of data set Y.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 13 — C

Choice C is correct. The median of a data set represented in a box plot is given by the vertical line within the box. In the given box plot, the vertical line within the box occurs at 5. Therefore, the median of this data set is 5.

Choice A is incorrect. This is the minimum value of the data set.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect. This is the maximum value of the data set.

▼ 14 — B

Choice B is correct. The median of a data set with an odd number of data values is defined as the middle value of the ordered list of values. The data set shown has nine values, so the median is the fifth value in the ordered list, which is 77.

Choice A is incorrect. This is the minimum value of the data set, not the median.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the mean of the data set, not the median.

▼ 15 — B

Choice B is correct. It's given that a data set consists of 173 colors and the bar graph shows the number of times each color appears in the data set. Therefore, for each color specified at the bottom of the bar, the frequency corresponds to the number of times that color

appears in the data set. The color that appears 70 times in the data set has a frequency of 70 on the bar graph. Since the bar with a frequency of 70 corresponds to green, green is the color that appears 70 times.

Choice A is incorrect. The color blue appears about 27 times, not 70 times.

Choice C is incorrect. The color red appears about 33 times, not 70 times.

Choice D is incorrect. The color yellow appears about 43 times, not 70 times.

▼ 16 — B

Choice B is correct. The median of the data summarized by a box plot is the value associated with the vertical line segment within the box. According to the box plot shown, this value is slightly greater than 7,000. Therefore, the closest value for the median number of acres is 7,067.

Choice A is incorrect. This is the value associated with the vertical line segment forming the left-hand side of the box. Choice C is incorrect. This value is greater than the value associated with the vertical line segment within the box. Choice D is incorrect. This is the value associated with the vertical line segment forming the right-hand side of the box.

▼ 17 — 40

The correct answer is 40. The height of each bar in the bar graph shown represents the number of cans collected by the group specified at the bottom of the bar. The bar for group 6 reaches a height of 40. Therefore, group 6 collected 40 cans.

▼ 18 — D

Choice D is correct. The range of a data set is the difference between the maximum and the minimum values in the set. The maximum value among the high temperatures in the table is 82°F and the minimum value is 80°F. Therefore, the range is 82°F – 80°F = 2°F.

Choice A is incorrect. The median of a data set is the middle value when the values in the set are ordered from least to greatest. Ordering the high temperatures this way gives the list 80, 81, 81, 81, 82. Therefore, the median high temperature is 81°F. Choice B is incorrect. The mean high temperature is $\frac{81+80+81+81+82}{5}=\frac{405}{5}=81$. Choice C is incorrect.

The mode is the value that occurs the greatest number of times. For the set of high

temperatures shown, 81 is the value that occurs 3 times, and therefore, 81°F is the mode of the high temperatures.

▼ 19 — C

Choice C is correct. The mean of a data set is found by dividing the sum of the values by the number of values. Therefore, the mean number of students who completed summer internships from Foothill High School is $\frac{87+80+75+76+70}{5}=\frac{388}{5}$, or 77.6. Similarly, the mean number from Valley High School is $\frac{44+54+65+76+82}{5}=\frac{321}{5}$, or 64.2. Thus, the mean number from Foothill High School is greater than the mean number from Valley High School. When a data set has an odd number of elements, the median can be found by ordering the values from least to greatest and determining the value in the middle. Since there are five values in each data set, the third value in each ordered list is the median. Therefore, the median number from Foothill High School is 76 and the median number from Valley High School is 65. Thus, the median number from Foothill High School is greater than the median number from Valley High School.

Choices A, B, and D are incorrect and may result from various misconceptions or miscalculations.

▼ 20 — 9

The correct answer is 9. The mean of a data set is the sum of the values in the data set divided by the number of values in the data set. It follows that the mean height, in centimeters, of these plants is the sum of the heights, in centimeters, of each plant, $6 + 10 + 13 + 2 + 15 + 22 + 10 + 4 + 4 + 4$, or 90, divided by the number of plants in the data set, 10. Therefore, the mean height, in centimeters, of these plants is $\frac{90}{10}$, or 9.

▼ 21 — B

Choice B is correct. The height of each bar in the bar graph given represents the number of students that voted for the activity specified at the bottom of the bar. The bar for activity 3 has a height that is between 35 and 40. In other words, the number of students that chose activity 3 is between 35 students and 40 students. Of the given choices, 39 is the only value between 35 and 40. Therefore, 39 students chose activity 3.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect. This is the number of students that chose activity 5, not activity 3.

Choice D is incorrect and may result from conceptual errors.

▼ 22 — 6

The correct answer is 6. It's given that the bar graph shows the frequency of each data value in a data set. The bar corresponding to the data value 2 has a height of 6. Therefore, the frequency of the data value 2 in this data set is 6.

▼ 23 — D

Choice D is correct. The mean of a data set is found by dividing the sum of the values in the data set by the number of values in the data set. Therefore, the mean of data set A is $\frac{1+2+3+4+5+6+7}{7}=\frac{28}{7}$, or 4. The mean of data set B is $\frac{1+1+2+2+3+3+4}{7}=\frac{16}{7}$, or approximately 2.2857. Therefore, the mean of data set A is greater than the mean of data set B.

Alternate approach: Data set A and data set B are both ordered from least to greatest value. Besides the first value in each data set, which is 1, each value in ordered data set B is less than the respective value in ordered data set A. Therefore, conceptually, the mean of data set A must be greater than the mean of data set B.

Choices A, B, and C are incorrect and may result from various misconceptions or miscalculations.

▼ 24 — C

Choice C is correct. The mean of a data set is found by dividing the sum of the values in the data set by the number of values in the data set. Therefore, the mean of data set A is $\frac{72+73+73+76+76}{5}$, which simplifies to 74. The mean of data set B is represented by the equation $\frac{61+64+74+85+x}{5}$, or $\frac{284+x}{5}$. It's given that the mean of data set A is equal to the mean of data set B. Therefore, the equation $74=\frac{284+x}{5}$ can be used to solve for x. Multiplying both sides of this equation by 5 yields $370=284+x$. Subtracting 284 from both sides of this equation yields $86=x$.

Choices A, B, and D are incorrect and may result from calculation errors.

▼ 25 — B

Choice B is correct. It's given that the bar graph shows the number of each type of monkey in a sanctuary. The bar representing the number of mandrills has a height of 5; therefore, there are 5 mandrills in the sanctuary. The bar representing vervets has a height of 11; therefore, there are 11 vervets in the sanctuary. Therefore, there are $11 - 5$, or 6, more vervets in this sanctuary than mandrills.

Choice A is incorrect. This is the number of vervets in the sanctuary.

Choice C is incorrect. This is the number of mandrills in the sanctuary.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 26 — 30

The correct answer is 30. The height of each bar in the bar graph shown represents the number of books collected by the group specified at the bottom of the bar. The bar for group 1 reaches a height of 30. Therefore, group 1 collected 30 books.

▼ 27 — D

Choice D is correct. It's given that the bar graph shows the number of students from one school who were born in either July, August, September, or October. The bar representing the number of students born in August has a height of 50; therefore, 50 students were born in August. The bar representing the number of students born in July has a height of 40; therefore, 40 students were born in July. Thus, there were $50 - 40$, or 10 more students born in August than in July.

Choice A is incorrect. This is the total number of students born in July and August.

Choice B is incorrect. This is the number of students born in August.

Choice C is incorrect. This is the number of students born in July.

▼ 28 — B

Choice B is correct. The numbers of defective lightbulbs found for the five trials are 4, 7, 1, 3, and 6, respectively. The mean is therefore $\frac{4+7+1+3+6}{5} = 4.2$.

Choice A is incorrect. This is the median number of defective lightbulbs for the five trials. Choice C is incorrect and may result from an arithmetic error. Choice D is incorrect and may result from mistaking the number of trials for the number of defective lightbulbs.

▼ 29 — B

Choice B is correct. If a data set contains an odd number of data values, the median is represented by the middle data value in the list when the data values are listed in ascending or descending order. Since the data set shown has 7 data values and is in ascending order, it follows that the median is the fourth data value in the list, or 8. If a data set contains an even number of data values, the median is between the two middle data values when the values are listed in ascending or descending order. Since each of the choices consists of a data set with 6 data values in ascending order, it follows that the median is between the third and fourth data value. The third and fourth data values in choice B are 8 and 8. Thus, choice B represents a data set with a median of 8. Since the median of the data set shown is 8 and choice B represents a data set with a median of 8, it follows that choice B represents a data set that has the same median as the data set shown.

Choice A is incorrect. This list represents a data set with a median of 6, not 8.

Choice C is incorrect. This list represents a data set with a median of 10, not 8.

Choice D is incorrect. This list represents a data set with a median of 10, not 8.

▼ 30 — A

Choice A is correct. It's given that the table shows an original data set of 5 values. It's also given that a sixth value is added to create a new data set. The new data set consists of the 5 values in the original data set and one additional value, 121. Since the additional value, 121, is less than any value in the original data set, the mean of the original data set is greater than the mean of the new data set.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 31 — B

Choice B is correct. Range is defined as the difference between the greatest and least values from a set of data. The greatest number of wheels made during a one-hour period was 24 wheels. The least number of wheels was 19. Hence, the range is $24-19=5$, or 5.0.

Choices A, C, and D are incorrect and may be the result of arithmetic errors or incorrectly identifying the greatest or least number of wheels made during a one-hour period.

▼ 32 — A

Choice A is correct. The average can be found by dividing the total number of boxes sold by the number of students, which is $\frac{90}{10}=9$.

Choices B, C, and D are incorrect. Each results from choosing measures that require the results of individual students, which are not given.

▼ 33 — C

Choice C is correct. The table gives the results of 200 people who were asked how often they see a movie in a theater. The table shows that 29 people responded “never” and 53 people responded “almost never.” Therefore, $29+53$, or 82, people responded either “never” or “almost never.”

Choice A is incorrect. This is the difference between the number of people who responded “almost never” and the number of people who responded “never.”

Choice B is incorrect. This is the number of people who responded “almost never” but doesn't include those who responded “never.”

Choice D is incorrect. This is the number of people who responded something other than “never” or “almost never,” rather than the number of people who responded either “never” or “almost never.”

▼ 34 — C

Choice C is correct. The table shows that for a certain region in 2016, the average number of store employees in warehouse stores was 365 and the average number of store employees in supermarkets was 130. Subtracting 130 from 365 yields $365-130$, or 235. Therefore, the average number of store employees was 235 greater in warehouse stores than in supermarkets.

Choice A is incorrect. For this region in 2016, this is how much greater the average number of store employees was in department stores than in supermarkets.

Choice B is incorrect. For this region in 2016, this is how much greater the average number of store employees was in warehouse stores than in department stores.

Choice D is incorrect. For this region in 2016, this is the sum of the average number of store employees in warehouse stores and in supermarkets.

▼ 35 — B

Choice B is correct. When a data set has an odd number of values, the median can be found by ordering the values from least to greatest and determining the value in the middle. Since the values are already presented in order from least to greatest and there are 7 values, the median is the fourth value in the list. Therefore, the median is 3.

Choice A is incorrect. This is the mode. Choice C is incorrect. This is the mean. Choice D is incorrect. This is the range.

▼ 36 — A

Choice A is correct. The mean low temperature can be calculated by finding the sum of the low temperatures for all the days shown in the table, 49 + 37 + 41 + 54 + 63 = 244, and then dividing the sum by the number of days the temperature was recorded, $244 \div 5 = 48.8$.

Choice B is incorrect. This may be the result of choosing the median rather than calculating the mean. Choices C and D are incorrect and may be the result of calculation errors.

▼ 37 — B

Choice B is correct. The height of each bar in the graph shown represents the number of containers that contain the number of walnuts specified at the bottom of the bar. The bar for 78 walnuts has a height of 7. Therefore, 7 of these containers of walnuts contain exactly 78 walnuts.

Choice A is incorrect. This is the number of containers that contain exactly 75 walnuts, not 78 walnuts.

Choice C is incorrect. This is the total number of containers of walnuts represented in the bar graph, not the number that contain exactly 78 walnuts.

Choice D is incorrect. This is the number of walnuts in a container that contains exactly 78 walnuts, not the number of containers that contain exactly 78 walnuts.

▼ 38 — C

Choice C is correct. Standard deviation is a measure of spread, so data sets with larger standard deviations tend to have larger spread. The standard deviation of the heights of the men in Group B is larger than the standard deviation of the heights of the men in Group A. Therefore, the heights of the men in Group B had a larger spread than the heights of the men in Group A.

Choice A is incorrect. If two data sets are identical, they will have equivalent means and equivalent standard deviations. Since the two data sets have different standard deviations, they cannot be identical. Choice B is incorrect. Without knowing the maximum value for each data set, it's impossible to know which group contained the tallest participant. Choice D is incorrect. Since the means of the two groups are equivalent, the medians could also be the same or could be different, but it's impossible to tell from the given information.

▼ 39 — 10

The correct answer is 10. The mean of a data set is calculated by dividing the sum of the data values by the number of data values in the data set. For this data set, the mean can be calculated as $\frac{6+8+16+4+17+26+8+5+5+5}{10}$, which is equivalent to $\frac{100}{10}$, or 10.

▼ 40 — A

Choice A is correct. The mode is the data value with the highest frequency. So for the data shown, the mode is 18. The median is the middle data value when the data values are sorted from least to greatest. Since there are 20 ages ordered, the median is the average of the two middle values, the 10th and 11th, which for these data are both 19. Therefore, the median is 19. The mean is the sum of the data values divided by the number of the data values. So for these data, the mean is

$$\frac{(18\times6)+(19\times5)+(20\times4)+(21\times2)+(22\times1)+(23\times1)+(30\times1)}{20}=20.$$

Since the mode is 18, the median is 19, and the mean is 20, $\text{mode} < \text{median} < \text{mean}$.

Choices B and D are incorrect because the mean is greater than the median. Choice C is incorrect because the median is greater than the mode.

Alternate approach: After determining the mode, 18, and the median, 19, it remains to determine whether the mean is less than 19 or more than 19. Because the mean is a balancing point, there is as much deviation below the mean as above the mean. It is possible to compare the data to 19 to determine the balance of deviation above and below the mean. There is a total deviation of only 6 below 19 (the 6 values of 18); however, the data value 30 alone deviates by 11 above 19. Thus the mean must be greater than 19.

▼ 41 — B

Choice B is correct. The median weight is found by ordering the horses' weights from least to greatest and then determining the middle value from this list of weights. Decreasing the value for the horse with the lowest weight doesn't affect the median since it's still the lowest value.

Choice A is incorrect. The mean is calculated by finding the sum of all the weights of the horses and then dividing by the number of horses. Decreasing one of the weights would decrease the sum and therefore decrease the mean. Choice C is incorrect. Range is the difference between the highest and lowest weights, so decreasing the lowest weight would increase the range. Choice D is incorrect. Standard deviation is calculated based on the mean weight of the horses. Decreasing one of the weights decreases the mean and therefore would affect the standard deviation.

▼ 42 — C

Choice C is correct. The median of a data set is the middle value when the data is in ascending or descending order. In ascending order, the gas prices are $3.609, $3.679, $3.699, $3.729, and $3.729. The middle number of this list is 3.699, so it follows that $3.699 is the median gas price.

Choice A is incorrect. When the gas prices are listed in ascending order, this value isn't the middle number. Choice B is incorrect. This value represents the mean gas price. Choice D is incorrect. This value represents both the mode and the maximum gas price.

▼ 43 — ?

The correct answer is 1.3. The median number of tourists is found by ordering the number of tourists from least to greatest and determining the middle value from this list. When the number of tourists in 2012 is ordered from least to greatest, the middle value, or the fifth number, is 46.4 million. When the number of tourists in 2013 is ordered from least to greatest, the middle value, or the fifth number, is 47.7 million. The difference between these two medians is $47.7 \text{ million} - 46.4 \text{ million} = 1.3 \text{ million}$. Note that 1.3 and 13/10 are examples of ways to enter a correct answer.

▼ 44 — B

Choice B is correct. The median of a set of numbers is the middle number when the values in the set are ordered from least to greatest. There are 33 fish, so in an ordered list of the weights, the 17th value would be the median weight. The 10 fish in tank A weigh the least, and these 10 weights would be the first 10 values on the ordered list. The 11 fish in tank B have the next set of higher weights, and so would be the 11th through 21st weights in the ordered list, which includes the median weight as the 17th value. The fish in tank B weigh at least 5 ounces but no more than 13 ounces; of the given choices, only 8 ounces falls within this range of values.

Choice A is incorrect. It's given that tank A has ten fish weighing less than 5 ounces. Since there are more than ten fish in tanks B and C combined, the median weight cannot be less than 5 ounces. Choice C and D are incorrect. It's given that tank C has twelve fish weighing more than 13 ounces. There are more than twelve fish in tanks A and B combined, so the median weight can't be more than 13 ounces.

▼ 45 — A

Choice A is correct. Standard deviation measures the spread of a given data set from its mean. In a data set with a smaller standard deviation, there are more values close to the mean. In a data set with a greater standard deviation, there are more values farther from the mean. The two histograms shown have the same scale on the horizontal axis. Therefore, their standard deviations can be compared by visually comparing the spreads of their histograms. The distribution summarized by each histogram is symmetric. Therefore, the mean of the data set for each histogram is a value in the middle bar of that histogram. The middle bar of each histogram has a value of at least 30 thousand units sold but less than 35 thousand units sold. Therefore, the mean of the data set for each histogram is at least 30 thousand and less than 35 thousand. The histogram for company A shows all the values in

that data set are close to the mean. For company B, the histogram shows there are fewer values close to the mean and more values farther from the mean. Therefore, the standard deviation of number of units sold for company A is less than the standard deviation of number of units sold for company B.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 46 — 19

The correct answer is 19. The minimum value of a data set is the least value in the data set. The frequency refers to the number of times a value occurs. The given table shows that for this data set, the value 19 occurs 7 times, the value 21 occurs 1 time, the value 23 occurs 7 times, and the value 25 occurs 4 times. Therefore, of the values 19, 21, 23, and 25 given in the data set, the minimum value of the data set is 19.

▼ 47 — A

Choice A is correct. The median is the middle value in a data set when the data are arranged in order from least to greatest. Since there are 15 values in data set A, the median is the 8th value. The 8th value is 23, so the median of data set A is 23. The range is found by subtracting the minimum value in a data set from the maximum value. The minimum value in data set A is 22 and the maximum value is 26. Therefore, the range of data set A is $26 - 22$, or 4. It's given that data set B is created by adding 46 to each of the values in data set A. Therefore, the 8th value in data set B is $23 + 46$, or 69, so the median of data set B is 69. The minimum value in data set B is $22 + 46$, or 68, and the maximum value is $26 + 46$, or 72. Therefore, the range of data set B is $72 - 68$, or 4. Since the median of data set A is 23, and the median of data set B is 69, the median of data set B is greater than the median of data set A. Since the ranges of data sets A and B are both 4, the range of data set B is equal to the range of data set A.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 48 — ?

The correct answer is 0. Any nine consecutive integers can be written as k, $k+1$, $k+2$, $k+3$, $k+4$, $k+5$, $k+6$, $k+7$, $k+8$. The mean of the integers is their sum divided by 9: $\frac{(k+k+1+k+2+...+k+8)}{9} = \frac{(9k+36)}{9}$, which simplifies to $k+4$. So $a=k+4$. Since there is an odd number of integers (nine), the median is the integer in the middle when all the integers are ordered from least to greatest: $k+4$. So $b=k+4$. Therefore, $|a-b| = |(k+4)-(k+4)|$, which is 0.

▼ 49 — ?

The correct answer is $\frac{1}{5}$. The mean of the list of numbers is found by dividing the sum of the numbers by the number of values in the list. Since there are 5 numbers in the list, the mean is $\frac{1}{5}$ of the sum of the numbers. Note that 1/5 and .2 are examples of ways to enter a correct answer.

▼ 50 — A

Choice A is correct. The mean of a data set is the sum of the values divided by the number of values. The mean of data set A is $\frac{45}{9}$, or 5. The mean of data set B is $\frac{145}{10}$, or 14.5. Thus, the means are different. The median of a data set is the middle value when the values are ordered from least to greatest. The medians of data sets A and B are both 5. Therefore, the medians are the same, so only the means are different.

Choices B, C, and D are incorrect and may result from conceptual or calculation errors.

▼ 51 — B

Choice B is correct. If a data set contains an even number of data values, when the data values are listed in ascending or descending order, the median is between the two middle values. The given data set contains 8 values. When listed in ascending order, the data set is $4, 4, 4, 5, 5, 6, 10, 18$ and the two middle values are 5 and 5. Since the two middle values are the same, the median must be 5.

Choice A is incorrect. This value is between the two middle values in the list shown, not the two middle values when the data values are listed in ascending or descending order.

Choice C is incorrect. This is the mean, not the median, of the data set.

Choice D is incorrect. This is the range, not the median, of the data set.

▼ 52 — 3630

The correct answer is $3{,}630$. The mean of a data set is the sum of the values in the data set divided by the number of values in the data set. The sum of the masses, in grams, of these alpine marmots is $4{,}010 + 4{,}010 + 3{,}030 + 4{,}050 + 3{,}050$, or $18{,}150$ grams. The number of alpine marmots in the data set is 5. Therefore, the mean mass, in grams, of these 5 alpine marmots is $\frac{18{,}150}{5}$, or $3{,}630$.

▼ 53 — 14

The correct answer is 14. The maximum value is the largest value in the data set. The frequency refers to the number of times a data value occurs. The given frequency table shows that for this data set, the data value 6 occurs three times, the data value 7 occurs three times, the data value 8 occurs eight times, the data value 9 occurs eight times, the data value 10 occurs nine times, the data value 11 occurs eleven times, the data value 12 occurs nine times, the data value 13 occurs zero times, and the data value 14 occurs six times. Therefore, the maximum data value in the data set is 14.

▼ 54 — C

Choice C is correct. If the values in a data set are ordered from least to greatest, the median of the data set will be the middle value. Since each data set in the choices is ordered and contains exactly 9 data values, the 5th value in each is the median. It follows that the median of the data set in choice C is 32. The sum of the positive differences between 32 and each of the values that are less than 32 is significantly smaller than the sum of the positive differences between 32 and each of the values that are greater than 32. If 32 were the mean, these sums would have been equal to each other. Therefore, the mean of this data set must be greater than 32. This can also be confirmed by calculating the mean as the sum of the values divided by the number of values in the data set:

$$\frac{2 + 4 + 8 + 16 + 32 + 64 + 128 + 256 + 512}{9} = 113\frac{5}{9}.$$

Choices A and B are incorrect. Each of the data sets in these choices is symmetric with respect to its median, so the mean and the median for each of these choices are equivalent. Choice D is incorrect. The median of this data set is 207. Since the sum of the positive differences between 207 and each of the values less than 207 is greater than the sum of

the positive differences between 207 and each value greater than 207 in this data set, the mean must be less than the median.

▼ 55 — A

Choice A is correct. If x is 6, then the five numbers in the given list are 15, 14, 18, 17, 6. The mean of these five numbers is the sum of all the values divided by the number of values, or $\frac{15+14+18+17+6}{5}=\frac{70}{5}=14$. The median of these five numbers can be found by ordering the numbers from least to greatest and determining the middle value. When ordered from least to greatest, the numbers in the given list are 6, 14, 15, 17, 18, and the middle value is 15. Since the mean is 14 and the median is 15, the mean and median aren't equal when x is 6.

Choices B, C, and D are incorrect. If any of these values is substituted for x, the mean and median of the data set would be equal.

▼ 56 — D

Choice D is correct. Standard deviation is a measure of the spread of the values in a data set from the mean of the data set, and the smaller the spread, the smaller the standard deviation. Each of the four choices represents a data set that is symmetric with respect to the value 54. Therefore, 54 is the mean of each of these data sets. All the values in the data set in choice D differ from 54 by either 0 or 1. In each of the other three data sets, the values in the data set, other than the value of 54, differ from 54 by at least 1. Therefore, the data set in choice D has the smallest spread and thus the smallest standard deviation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 57 — B

Choice B is correct. Standard deviation is a measure of the spread of a data set from its mean. The dot plot for class A and the dot plot for class B have the same shape. Thus, the frequency distributions for both class A and class B are the same. Since both class A and class B have the same frequency distribution of glue sticks brought in by each student, it

follows that both class A and class B have the same spread of the number of glue sticks brought in by each student from their respective means. Therefore, the standard deviation of the number of glue sticks brought in by each student for class A is equal to the standard deviation of the number of glue sticks brought in by each student for class B.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 58 — 29

The correct answer is 29. The range of a data set is the difference between its maximum value and its minimum value. For the data set shown, the maximum score is 52 and the minimum score is 23. The difference between those scores is 52 - 23, or 29. Therefore, the range of the 7 scores shown is 29.

▼ 59 — C

Choice C is correct. If the mean score of 8 players is 14.5, then the total of all 8 scores is $14.5 \times 8 = 116$. If the mean of 7 scores is 12, then the total of all 7 scores is $12 \times 7 = 84$. Since the set of 7 scores was made by removing the highest score of the set of 8 scores, then the difference between the total of all 8 scores and the total of all 7 scores is equal to the removed score: $116 - 84 = 32$.

Choice A is incorrect because if 20 is removed from the group of 8 scores, then the mean score of the remaining 7 players is $\frac{(14.5 \times 8) - 20}{7}$ is approximately 13.71, not 12. Choice B is incorrect because if 24 is removed from the group of 8 scores, then the mean score of the remaining 7 players is $\frac{(14.5 \times 8) - 24}{7}$ is approximately 13.14, not 12. Choice D is incorrect because if 36 is removed from the group of 8 scores, then the mean score of the remaining 7 players is $\frac{(14.5 \times 8) - 36}{7}$ or approximately 11.43, not 12.

▼ 60 — 44

The correct answer is 44. The mean of a data set is computed by dividing the sum of the values in the data set by the number of values in the data set. It's given that data set A consists of the heights of 75 buildings and has a mean of 32 meters. This can be represented by the equation $\frac{x}{75} = 32$, where x represents the sum of the heights of the buildings, in meters, in data set A. Multiplying both sides of this equation by 75 yields $x = 7532$, or $x = 2{,}400$ meters. Therefore, the sum of the heights of the buildings in data set A is 2,400 meters. It's also given that data set B consists of the heights of 50 buildings and has a mean of 62 meters. This can be represented by the equation $\frac{y}{50} = 62$, where y represents the sum of the heights of the buildings, in meters, in data set B. Multiplying both sides of this equation by 50 yields $y = 5062$, or $y = 3{,}100$ meters. Therefore, the sum of the heights of the buildings in data set B is 3,100 meters. Since it's given that data set C consists of the heights of the 125 buildings from data sets A and B, it follows that the mean of data set C is the sum of the heights of the buildings, in meters, in data sets A and B divided by the number of buildings represented in data sets A and B, or $\frac{2{,}400 + 3{,}100}{125}$, which is equivalent to 44 meters. Therefore, the mean, in meters, of data set C is 44.

▼ 61 — D

Choice D is correct. The mean amount of time that the 20 employees worked for the company is 6.7 years. This means that the total number of years all 20 employees worked for the company is (6.7)(20) = 134 years. After the employee left, the mean amount of time that the remaining 19 employees worked for the company is 6.25 years. Therefore, the total number of years all 19 employees worked for the company is (6.25)(19) = 118.75 years. It follows that the number of years that the employee who left had worked for the company is 134 – 118.75 = 15.25 years.

Choice A is incorrect; this is the change in the mean, which isn't the same as the amount of time worked by the employee who left. Choice B is incorrect and likely results from making the assumption that there were still 20 employees, rather than 19, at the company after the employee left and then subtracting the original mean of 6.7 from that result. Choice C is incorrect and likely results from making the assumption that there were still 20 employees, rather than 19, at the company after the employee left.

▼ 62 — 4

The correct answer is 4. It's given that the two boxplots show the distribution of number of books read over the summer by the students in two different English classes. In a boxplot, the first vertical line represents the minimum value of the data set and the last vertical line

represents the maximum value of the data set. The range of a data set is the difference between its maximum value and its minimum value. In class A, the maximum number of books read is 5 and the minimum number of books read is 0. The difference between those values is $5 - 0$, or 5. Therefore, the range of the number of books read in class A is 5. In class B, the maximum number of books read is 10 and the minimum number of books read is 1. The difference between those values is $10 - 1$, or 9. Therefore, the range of the number of books read in class B is 9. To find the positive difference between the ranges of the number of books read for the two classes, the smaller range must be subtracted from the larger range. Therefore, the positive difference between the ranges of number of books read over the summer for the two classes is $9 - 5$, or 4.

▼ 63 — B

Choice B is correct. The median of a set of numbers is the middle value of the set values when ordered from least to greatest. If the percents in the table are ordered from least to greatest, the middle value is 27.9%. The difference between 27.9% and 26.95% is 0.95%.

Choice A is incorrect and may be the result of calculation errors or not finding the median of the data in the table correctly. Choice C is incorrect and may be the result of finding the mean instead of the median. Choice D is incorrect and may be the result of using the middle value of the unordered list.

▼ 64 — ?

The correct answer is 2.6. Since the mean of a set of numbers can be found by adding the numbers together and dividing by how many numbers there are in the set, the mean mass, in kilograms, of the rocks Andrew collected is $\frac{2.4+2.5+3.6+3.1+2.5+2.7}{6} = \frac{16.8}{6}$, or 2.8. Since the mean mass of the rocks Maria collected is 0.1 kilogram greater than the mean mass of rocks Andrew collected, the mean mass of the rocks Maria collected is $2.8+0.1=2.9$ kilograms. The value of x can be found by writing an equation for finding the mean: $\frac{x+3.1+2.7+2.9+3.3+2.8}{6} = 2.9$. Solving this equation gives $x = 2.6$. Note that 2.6 and 13/5 are examples of ways to enter a correct answer.

▼ 65 — 52

The correct answer is 52. The mean of a data set is calculated by dividing the sum of the data values by the number of values. It's given that data set A consists of 10 values, 9 of which are shown. Let x represent the 10th data value in data set A, which isn't shown. The mean of data set A can be found using the expression $\frac{43 + 45 + 44 + 43 + 38 + 39 + 40 + 46 + 40 + x}{10}$, or $\frac{378 + x}{10}$. It's given that the mean of the 9 values shown is 42 and that the mean of all 10 numbers is greater than 42. Consequently, the 10th data value, x, is larger than 42. It's also given that the data values in data set A are positive integers less than 60. Thus, $42 < x < 60$. Finally, it's given that the mean of data set A is an integer. This means that the sum of the 10 data values, $378 + x$, is divisible by 10. Thus, $378 + x$ must have a ones digit of 0. It follows that x must have a ones digit of 2. Since $42 < x < 60$ and x has a ones digit of 2, the only possible value of x is 52. Since 52 is larger than any of the integers shown, the largest integer from data set A is 52.

▼ 66 — B

Choice B is correct. The mean of a data set is the sum of the values in the data set divided by the number of values in the data set. The new data set consists of the weights of the 71 tortoises in the original data set and one additional weight, 39. Since the additional weight, 39, is greater than any of the values in the original data set, the mean of the new data set is greater than the mean of the original data set. If a data set contains an odd number of data values, the median is represented by the middle data value in the list when the data values are listed in ascending or descending order. Since the original data set consists of the weights of 71 tortoises and is in ascending order, the median of the original data set is represented by the middle value, or the 36th value. Based on the frequencies shown in the table, the 36th value in this data set is 17. If a data set contains an even number of data values, the median is between the two middle data values when the values are listed in ascending or descending order. Since the new data set consists of the weights of 72 tortoises, the median of the new data set is between the 36th and 37th data values when the values are arranged in ascending order. To keep the data in ascending order, the additional value of 39 would be placed at the bottom of the frequency table with a frequency of 1. Therefore, based on the frequencies in the table, the 36th and 37th values in the new data set are both 17. It follows that the median of the new data set is 17, which is the same as the median of the original data set. Therefore, the mean of the new data set is greater than the mean of the original data set, and the medians of the two data sets are equal.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 67 — D

Choice D is correct. It's given that data set F consists of 55 integers between 170 and 290 and data set G consists of all the integers in data set F as well as the integer 10. Since the integer 10 is less than all the integers in data set F, the mean of data set G must be less than the mean of data set F. Thus, the mean of data set F isn't less than the mean of data set G. When a data set is in ascending order, the median is between the two middle values when there is an even number of values and the median is the middle value when there is an odd number of values. It follows that the median of data set F is either greater than or equal to the median of data set G. Therefore, the median of data set F isn't less than the median of data set G. Thus, neither the mean nor the median must be less for data set F than for data set G.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 68 — D

Choice D is correct. When a data set has an odd number of elements, the median can be found by ordering the values from least to greatest and determining the middle value. Out of the 27 different numbers in this data set, 13 numbers are below the median, one number is exactly 33, and 13 numbers are above the median. When 7 is subtracted from each number below the median and added to each number above the median, the data spread out from the median. Since the median of this data set, 33, is equivalent to the mean of the data set, the data also spread out from the mean. Since standard deviation is a measure of how spread out the data are from the mean, a greater spread from the mean indicates an increased standard deviation.

Choice A is incorrect. All the numbers less than the median decrease and all the numbers greater than the median increase, but the median itself doesn’t change. Choices B and C are incorrect. The mean of a data set is found by dividing the sum of the values by the number of values. The net change from subtracting 7 from 13 numbers and adding 7 to 13 numbers is zero. Therefore, neither the mean nor the sum of the numbers changes.

▼ 69 — B

Choice B is correct. The mean of a data set is the sum of the values divided by the number of values. It's given that a new player earns 18 points playing the game, and this number of points is added to data set A to create data set B with 51 values. The histogram shows that all the values in data set A are greater than 18. Since the additional number of points, 18, is less than any of the values in the original data set, the mean number of points per player for data set B is less than the mean number of points per player for data set A. The median of a data set with 50 values is between the 25th and 26th values when the values are listed in ascending order. The median of a data set with 51 values is the 26th value when the values are listed in ascending order. Therefore, the median of data set B is the 26th value in data set B, which is the 25th value in data set A. The histogram shows that the 25th and 26th values in data set A are both between 50 and 60 points. Therefore, the 25th and 26th values in data set A could be equal, in which case the median of data set A would be equal to both the 25th and 26th values in data set A. Then, the median number of points per player for data set A would be equal to the median number of points per player for data set B. Therefore, only statement II must be true.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 70 — A

Choice A is correct. Since the frequencies of values less than the middle value, 3, are the same as the frequencies of the values greater than 3, the set of data has a symmetric distribution. When a set of data has a symmetric distribution, the mean and median values are equal. Therefore, the mean is 0 greater than the median.

Choices B, C, and D are incorrect and may result from misinterpreting the set of data.

▼ 71 — 41

The correct answer is 41. The mean of a data set is computed by dividing the sum of the values in the data set by the number of values in the data set. It’s given that data set A consists of the heights of 75 objects and has a mean of 25 meters. This can be represented

by the equation $\frac{x}{75} = 25$, where x represents the sum of the heights of the objects, in meters, in data set A. Multiplying both sides of this equation by 75 yields $x = 7525$, or $x = 1{,}875$ meters. Therefore, the sum of the heights of the objects in data set A is $1{,}875$ meters. It's also given that data set B consists of the heights of 50 objects and has a mean of 65 meters. This can be represented by the equation $\frac{y}{50} = 65$, where y represents the sum of the heights of the objects, in meters, in data set B. Multiplying both sides of this equation by 50 yields $y = 5065$, or $y = 3{,}250$ meters. Therefore, the sum of the heights of the objects in data set B is $3{,}250$ meters. Since it's given that data set C consists of the heights of the 125 objects from data sets A and B, it follows that the mean of data set C is the sum of the heights of the objects, in meters, in data sets A and B divided by the number of objects represented in data sets A and B, or $\frac{1{,}875 + 3{,}250}{125}$, which is equivalent to 41 meters. Therefore, the mean, in meters, of data set C is 41.

▼ 72 — C

Choice C is correct. The median of a data set with an odd number of values, in ascending or descending order, is the middle value of the data set, and the range of a data set is the positive difference between the maximum and minimum values in the data set. Since the dot plot shown gives the values in data set A in ascending order and there are 15 values in the data set, the eighth value in data set A, 23, is the median. The maximum value in data set A is 26 and the minimum value is 22, so the range of data set A is $26 - 22$, or 4. It's given that data set B is created by adding 56 to each of the values in data set A. Increasing each of the 15 values in data set A by 56 will also increase its median value by 56 making the median of data set B 79. Increasing each value of data set A by 56 does not change the range, since the maximum value of data set B is $26 + 56$, or 82, and the minimum value is $22 + 56$, or 78, making the range of data set B $82 - 78$, or 4. Therefore, the median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 73 — ?

The correct answer is 1. When there are an odd number of values in a data set, the median of the data set is the middle number when the data values are ordered from least to

greatest. The scores for group A, ordered from least to greatest, are 2, 3, 4, 4, 5, 6, 6, 6, and 9. The median of the scores for group A is therefore 5. The scores for group B, ordered from least to greatest, are 5, 5, 5, 5, 6, 6, 8, 8, 9, 10, and 10. The median of the scores for group B is therefore 6. The median score for group B is $6-5=1$ more than the median score for group A.

▼ 74 — A

Choice A is correct. The tables in choices B, C, and D each represent a data set where the values 80 and 90 have the same frequency and the values 70 and 100 have the same frequency. It follows that each of these data sets is symmetric around the value halfway between 80 and 90, or 85. When a data set is symmetric around a value, that value is the mean of the data set. Therefore, the data sets represented by the tables in choices B, C, and D each have a mean of 85. The table in choice A represents a data set where the value 90 has a greater frequency than the value 80 and the value 100 has a greater frequency than the value 70. It follows that this data set has a mean greater than 85. Therefore, of the given choices, choice A represents the data set with the greatest mean.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 75 — B

Choice B is correct. The mean of a data set is calculated by dividing the sum of the values in the data set by the number of values in the data set. It's given that the mean length of the 240 gray seals from Muskeget Island was 88 inches. This can be represented by the equation $\frac{x}{240} = 88$, where x represents the sum of the lengths, in inches, of the 240 gray seals from Muskeget Island. Multiplying both sides of this equation by 240 yields $x = (240)(88)$, or $x = 21{,}120$ inches. It's also given that the mean length of the 120 gray seals from Sable Island was 94 inches. This can be represented by the equation $\frac{y}{120} = 94$, where y represents the sum of the lengths, in inches, of the 120 gray seals from Sable Island. Multiplying both sides of this equation by 120 yields $y = (120)(94)$, or $y = 11{,}280$ inches. Therefore, the sum of the lengths, in inches, of all 360 gray seals is $21{,}120 + 11{,}280$, or $32{,}400$. Dividing this sum by 360 yields $\frac{32{,}400}{360}$, or 90 inches. Therefore, the mean length of all 360 gray seals the scientist measured for this study is 90 inches.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the mean length of 88 inches and 94 inches, not the mean length of all 360 gray seals the scientist measured for this study.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 76 — A

Choice A is correct. The mean value of a data set is the sum of the values of the data set divided by the number of values in the data set. When a data set is represented in a frequency table, the sum of the values in the data set is the sum of the products of each value and its frequency. For data set A, the sum of products of each value and its frequency is $302 + 344 + 385 + 427 + 469$, or $1{,}094$. It's given that there are 27 values in data set A. Therefore, the mean of data set A is $\frac{1{,}094}{27}$, or approximately 40.52. Similarly, the mean of data B is $\frac{958}{27}$, or approximately 35.48. Therefore, the mean of data set A is greater than the mean of data set B.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 77 — C

Choice C is correct. It's given that the data set containing a, 26, 29, b, 31, 47, and c is listed in ascending order and has a mean of 36, a median of 29, and a range of 72. Since there are 7 values in the given data set, it follows that the median is the value in the fourth position. It's given that b is in the fourth position in the given data set, so it follows that $b = 29$. Since the mean of the data set is 36, it follows that $\frac{(a + 26 + 29 + 29 + 31 + 47 + c)}{7} = 36$. Multiplying both sides of this equation by 7 yields $a + 26 + 29 + 29 + 31 + 47 + c = 252$, which is equivalent to $a + c + 162 = 252$. Subtracting 162 from both sides of this equation yields $a + c = 90$. The range is the difference between the greatest and least values in a data set. Since c is the greatest value and a is the least value in the given data set, it follows that $c - a = 72$, or $-a + c = 72$. Adding the equations $a + c = 90$ and $-a + c = 72$ yields $2c = 162$. Dividing both sides of this equation by 2 yields $c = 81$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 78 — C

Choice C is correct. The median of a data set represented in a box plot is represented by the vertical line within the box. It follows that the median mass of the gazelles in group 1 is 25 kilograms, and the median mass of the gazelles in group 2 is 24 kilograms. Since 25 kilograms is greater than 24 kilograms, the median mass of group 1 is greater than the median mass of group 2.

Choice A is incorrect. The mean mass of each of the two groups cannot be determined from the box plots.

Choice B is incorrect. The mean mass of each of the two groups cannot be determined from the box plots.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 79 — A

Choice A is correct. The median of a data set with an odd number of values that are in ascending or descending order is the middle value of the data set. Since the distribution of the values of both data set A and data set B form symmetric dot plots, and each data set has an odd number of values, it follows that the median is given by the middle value in each of the dot plots. Thus, the median of data set A is 13, and the median of data set B is 13. Therefore, statement I is true. Data set A and data set B have the same frequency for each of the values 11, 12, 14, and 15. Data set A has a frequency of 1 for values 10 and 16, whereas data set B has a frequency of 2 for values 10 and 16. Standard deviation is a measure of the spread of a data set; it is larger when there are more values further from the mean, and smaller when there are more values closer to the mean. Since both distributions are symmetric with an odd number of values, the mean of each data set is equal to its median. Thus, each data set has a mean of 13. Since more of the values in data set A are closer to 13 than data set B, it follows that data set A has a smaller standard deviation than data set B. Thus, statement II is false. Therefore, only statement I must be true.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 80 — B

Choice B is correct. The mean of a data set is the sum of the values of the data set divided by the total number of values in the data set. When a data set is represented in a frequency table, the sum of the values in the data set is the sum of the products of each value and its frequency. For data set Q, the sum of the products of each value and its frequency is $110a + 30b + 0c$, or $110a + 30b$, and the number of values is $110 + 30 + 0$, or 140. Therefore, the mean of data set Q is $\frac{110a + 30b}{140}$. For data set R, the sum of the products of each value and its frequency is $0a + 3b + 11c$, or $3b + 11c$, and the number of values is $0 + 3 + 11$, or 14. Therefore, the mean of data set R is $\frac{3b + 11c}{14}$. Since a, b, and c are positive, consecutive integers and $a < b < c$, it follows that $b = a + 1$ and $c = a + 2$. Substituting $a + 1$ for b in the mean of data set Q yields $\frac{110a + 30(a + 1)}{140}$, or $\frac{110a + 30a + 30}{140}$, which is equivalent to $\frac{140a + 30}{140}$, or $a + \frac{3}{14}$. Substituting $a + 1$ for b and $a + 2$ for c in the mean of data set R yields $\frac{3(a + 1) + 11(a + 2)}{14}$, or $\frac{3a + 3 + 11a + 2(11)}{14}$, which is equivalent to $\frac{14a + 3 + 2(11)}{14}$, or $a + \frac{3}{14} + \frac{2(11)}{14}$. Subtracting the mean of data set Q from the mean of data set R yields $(a + \frac{3}{14} + \frac{2(11)}{14}) - (a + \frac{3}{14})$, or $\frac{2(11)}{14}$. Therefore, the mean of data set R is $\frac{2(11)}{14}$ greater than the mean of data set Q.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 81 — C

Choice C is correct. It's given that the dot plot represents a data set of the number of bursts for 13 eruptions of a steam vent. The median of a data set with an odd number of elements is the middle element when the elements are in numerical order. For 13 elements in numerical order, this is the 7th element. For this data set, the first 4 elements have a value of 1, and the next 5 elements have a value of 2. Thus, the 7th element in the ordered data set is 2 and the median number of bursts for the original data set is 2. If an additional eruption with 11 bursts is added to this data set to create a new data set of 14 eruptions, the median of the new data set will be between the 7th and 8th elements in the ordered set, which will also be 2. Therefore, the median number of bursts for the new data set will be the same as the median number of bursts for the original data set. The mean number of bursts for the original data set is found by adding the values of all 13 elements and dividing that sum by the number of elements, 13. Since the data is shown in a dot plot, the sum of the values of the elements can be found by multiplying each element's value by its frequency: $14 + 25 + 32 + 41 + 51$, or 29. Therefore, the mean number of bursts for the original data set

is $\frac{29}{13}$. If an additional eruption with 11 bursts is added to this data set to create a new data set of 14 bursts, the mean number of bursts for the new data set is $\frac{29+11}{14}$, or $\frac{40}{14}$. Since $\frac{40}{14} > \frac{29}{13}$, the mean number of bursts for the new data set is greater than the mean number of bursts for the original data set. Therefore, of the median number of bursts and the mean number of bursts, only the mean number of bursts is greater for the new data set than for the original data set.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 82 — D

Choice D is correct. If a data set contains an odd number of data values, the median is represented by the middle data value in the list when the data values are listed in ascending or descending order. Since the numbers of employees are given as ranges of values rather than specific values, it's only possible to determine the range in which the median falls, rather than the exact median. Since there are 17 restaurants included in the data set and the numbers of employees are listed in ascending order, it follows that the median number of employees will be represented by the ninth restaurant in the list. Since the first 2 restaurants each have 2 to 7 employees, numbers of employees in the 2 to 7 range would be represented by the first and second restaurants in the list. The next 4 restaurants each have 8 to 13 employees. Therefore, numbers of employees in the 8 to 13 range will be represented by the third through sixth restaurants in the list. The next 2 restaurants each have 14 to 19 employees. Therefore, numbers of employees in the 14 to 19 range will be represented by the seventh and eighth restaurants in the list. Since the next 7 restaurants each have 20 to 25 employees, numbers of employees in the 20 to 25 range will be represented by the ninth through fifteenth restaurants in the list. This means that the ninth restaurant in the list, which has the median number of employees for the restaurants in this town, has a number of employees in the 20 to 25 range. Of the given choices, the only number of employees in the 20 to 25 range is 21.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the position of the median in the list, not the value of the median.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 83 — C

Choice C is correct. It's given that for data set A, the table summarizes the number of deliveries received by an office each day over a period of 11 days. If a data set contains an odd number of data values, the median is represented by the middle data value in the list when the data values are listed in ascending or descending order. Since data set A contains 11 data values in ascending order, the median of data set A is the 6th value. Based on the number of days for each value shown in the table, the 6th value is 4, so the median of data set A is 4. It's also given that the data value 13 was recorded in error and is removed from data set A to create data set B. It follows that data set B contains 10 data values in ascending order. If a data set contains an even number of data values, the median is between the two middle data values when the values are listed in ascending or descending order. It follows that the median of data set B is between the 5th and 6th values. Based on the number of days for each value shown in the table, the 5th and 6th values are both 4, so the median of data set B is 4. Therefore, the median of data set B is equal to the median of data set A.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 84 — B

Choice B is correct. The histograms shown have the same shape, but data set A contains values between 20 and 60 and data set B contains values between 10 and 50. Thus, the mean of data set A is greater than the mean of data set B. Therefore, the smallest possible difference between the mean of data set A and the mean of data set B is the difference between the smallest possible mean of data set A and the greatest possible mean of data set B. In data set A, since there are 3 integers in the interval greater than or equal to 20 but less than 30, 4 integers greater than or equal to 30 but less than 40, 7 integers greater than or equal to 40 but less than 50, and 9 integers greater than or equal to 50 but less than 60, the smallest possible mean for data set A is $\frac{3 \cdot 20 + 4 \cdot 30 + 7 \cdot 40 + 9 \cdot 50}{23}$. In data set B, since there are 3 integers greater than or equal to 10 but less than 20, 4 integers greater than or equal to 20 but less than 30, 7 integers greater than or equal to 30 but less than 40, and 9 integers greater than or equal to 40 but less than 50, the largest possible mean for data set B is $\frac{3 \cdot 19 + 4 \cdot 29 + 7 \cdot 39 + 9 \cdot 49}{23}$. Therefore, the smallest possible difference between the mean of data set A and the mean of data set B is $\frac{3 \cdot 20 + 4 \cdot 30 + 7 \cdot 40 + 9 \cdot 50}{23} - \frac{3 \cdot 19 + 4 \cdot 29 + 7 \cdot 39 + 9 \cdot 49}{23}$, which is

equivalent to $\frac{3 \cdot 20 - 3 \cdot 19 + 4 \cdot 30 - 4 \cdot 29 + 7 \cdot 40 - 7 \cdot 39 + 9 \cdot 50 - 9 \cdot 49}{23}$. This expression can be rewritten as $\frac{320 - 19 + 430 - 29 + 740 - 39 + 950 - 49}{23}$, or $\frac{23}{23}$, which is equal to 1. Therefore, the smallest possible difference between the mean of data set A and the mean of data set B is 1.

Choice A is incorrect. This is the smallest possible difference between the ranges, not the means, of the data sets.

Choice C is incorrect. This is the difference between the greatest possible mean, not the smallest possible mean, of data set A and the greatest possible mean of data set B.

Choice D is incorrect. This is the smallest possible difference between the sum of the values in data set A and the sum of the values in data set B, not the smallest possible difference between the means.